R2.3 How far? The extent of chemical change

The big picture

Guiding question(s)

How can the extent of a reversible reaction be influenced?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

In most of the chemical reactions you have studied so far, the assumption has been that the reaction goes to completion. This is taken to mean that the reaction only goes in the forward direction and all the reactants have been converted to products. In this subtopic you will learn that chemical reactions can actually go in both directions, in the forward and reverse directions. These reactions are known as reversible reactions (**Figure 1**).

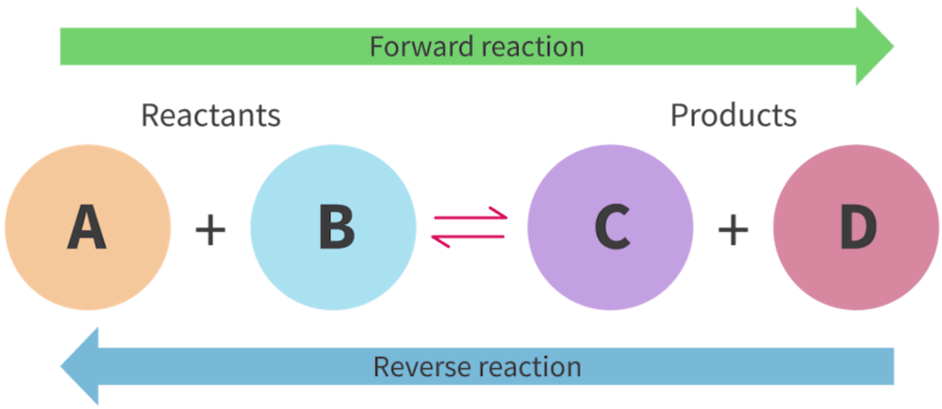

Figure 1. Reversible chemical reactions can proceed in both directions.

More information for figure 1

How far a reaction proceeds in the forward or reverse direction is known as the extent of reaction. The extent of reaction is of particular importance for industrial processes involving reversible reactions. Two notable examples are the Haber process that produces ammonia and the Contact process that produces sulfuric acid. Both of these reactions are carried out under carefully controlled conditions to maximise the yield of the reaction.

The ammonia produced in the Haber process is used to make many important chemicals as shown in **Figure 2**.

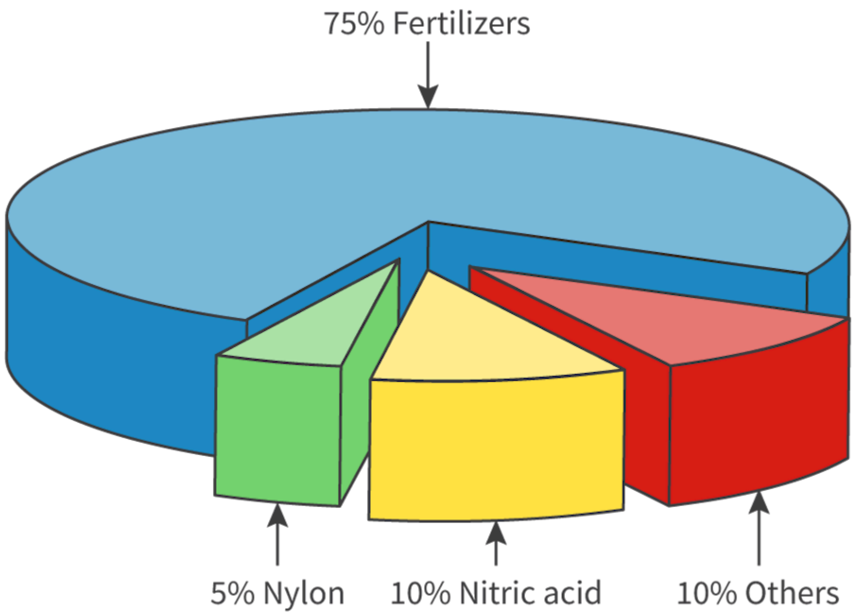

Figure 2. The Haber process produces ammonia which is used to produce a variety of important chemicals — the most important being fertiliser which is used to grow crops.

More information for figure 2

As you work your way through this section, keep in mind the reversible nature of chemical reactions as you learn about the factors that affect the extent of reversible reactions.

Think about why some reactions proceed more in one direction than the other and what factors influence the extent of a chemical reaction.

Prior learning

Before you study this subtopic make sure that you understand the following:

- The kinetic molecular theory is a model to explain physical properties of matter (solids, liquids and gases) and changes of state section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/).

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- The relationship between the pressure, volume, temperature and amount of an ideal gas is shown in the ideal gas equation $PV = nRT$ section S1.5.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ideal-gas-equation-id-44353/).
- Catalysts increase the rate of reaction by providing an alternative reaction pathway with lower E_a section R2.2.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/).